Abel Lawrence Peirson

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Education

Stanford University — *Ph.D. in Physics* *2017 - Mar 2023*
- Thesis: Relativistic Jets in the Era of X-ray Polarimetry.
- GPA: 4.0 (in required coursework).

University of Oxford — *MPhys in Physics* *2013 - 2017*
- First Class Honours with Distinction – top 10% of graduating class.
- College: Christ Church

Honors & Awards

Stanford Data Science Scholar ($100k+) — Stanford, CA *2021*

Future Investigator in NASA Earth and Space Science and Technology FINESST ($160k+) — Stanford, CA *2019*

Hooke & Roach Prizes: top STEM student, Christ Church — Oxford, UK *2017*

Oxford International Strategy Scholarship (CERN) — Oxford, UK *2016*

Gold, British Physics Olympiad — London, UK *2013*

Featured in: NASA Blog (2022) — Hogg's Research (2022) — NVIDIA AI Podcast, ep. 68 (2018) — TechCrunch (2018)

Experience

Citadel GQS — New York / Miami *2023 - Present*
Quantitative Researcher
- Alpha research: sequence modelling and structured deep learning.

Kavli Institute for Particle Astrophysics and Cosmology — Stanford University, CA *July 2018 - Mar 2023*
*with **Prof. Roger Romani***
- State-of-the-art X-ray polarization recovery using simulation-based inference.
- $>30\%$ improvement in NASA IXPE polarization sensitivity (code adopted as official data analysis pipeline).
- Quadratic program + nested sampling approach to fit gravitational microlenses.
- Testable (and fittable) emission models to explain observed polarization in relativistic plasma jets.

Google Brain — Mountain View, CA *June - Dec 2022*
Student Researcher
- Optimization for deep learning, NTK-NNPK correspondence.
- Layerwise Bregman exponential families for preconditioning SGD with local Fisher approximations.
- MLCommons algorithmic efficiency benchmarking.
- Mentors: Rohan Anil / Ehsan Amid / Prof. Manfred K. Warmuth.

G-Research — London, UK *June - Sep 2021*
Quantitative Researcher Intern
- Algorithmic trading detection.

Peirson & Freedman — Stanford, CA *Aug 2018 - 2019*
Co-founder
- Built iOS app ***Dank Learning*** that uses RNNs to generate memes. [Webpage]

Wu Tsai Neurosciences Institute — Stanford University, CA *Mar 2018 - Jun 2018*
*with **Prof. Shaul Druckmann***
- Biologically inspired recurrent neural network to reproduce 2D path integration in the Drosophila brain.

NeuroAI Lab — Stanford University, CA *Dec 2017 - Mar 2018*
*with **Prof. Dan Yamins***

- Unsupervised learning: shear-transformation pretraining does not improve image classification.

Department of Physics — University of Oxford, UK *Sep 2016 - May 2017*
*with **Prof. Garret Cotter***

- Placed limits on whether the Cherenkov Telescope Array will constrain the existence of axion-like particles.

CLIC Test Facility — CERN, Switzerland *June - Aug 2016*
*with **Prof. Philip Burrows***

- Improved CTF3's Quadrupole scan and reduced uncertainty in the beam energy spread.

Presentations

— Invited —

IXPE: Science so far, *IAUS 375: The Multimessenger Chakra of Blazar Jets*, Dec 2022

X-ray Polarimetry: A data intensive science, *Stanford Data Science Seminar*, 2022

Tests of gravitational milli-lensing in the blazar PKS 1413+135, *Max Planck Institute for Radio Astronomy*, 2022

Optimal Signal Extraction for IXPE and an Application to Blazars, *Naval Research Laboratory Colloquium*, 2021

Towards Optimal Signal Extraction for IXPE, *Third Science Collaboration Meeting (SCM03)*, 2021

The Polarization Behavior of Synchrotron Self-Compton Emission in Blazars, *Understanding the Multiwavelength Blazar Variability - Workshop, Stanford*, 2019

AI in Design, *Used Future: Symposium by Current Obsession*, Pratt Institute NY, 2018

— Contributed —

Relativistic Jets in the Era of X-ray Polarimetry, *Columbia THEA Seminar*, Nov 2022

Fishy: Local Fisher Approximation via Shampoo, *Google Brain Weekly*, Nov 2022

Blazar rotations with IXPE, *KIPAC Tea*, Oct 2022

A Geometric Model for Blazar Rotations, *Meeting of the European Astronomical Society*, 2020

Telescope & Computing Allocations

ALMA — *High-frequency radio observations of IXPE targets*, 7-14 day (Co-I) *2021*
Nordic Optical Telescope — *Multi-band polarization observations of IXPE targets*, 87ks (Co-I) *2021*
NuSTAR — *Multi-Energy X-ray observations of IXPE blazar targets*, 140ks (Co-I) *2021*
Swift — *Monitoring IXPE blazar targets with Swift*, 240ks (Co-I) *2020*
Google Cloud Platform — *Parametric Density Estimation with Uncertainty using Deep Ensembles*, $1000 *2020*
XMM-Newton — *Exploring the Synchro-Compton transition in CGRaBS J0211+1051*, 57ks (Co-I) *2019*

Open Source (★1000+)

Python — C/C++ — JAX — PyTorch — TensorFlow

MLCommons Algorithmic Efficiency: Benchmark and competition for neural network training. [Code][Webpage]
ixpeobssim: Simulation and analysis framework for the Imaging X-ray Polarimetry Explorer. [Code][Webpage]
MulensModel: Python package for modelling gravitational microlensing events. [Code][Paper][Webpage]
SSCpol: Polarized relativistic jet simulation in C with Python wrapper. [Code][Paper]
Dank Learning: 'Show and Tell' image captioning for meme generation in TensorFlow. [Code][Paper][Webpage]

Outreach and Service

Wonderfest — Bay Area, CA *June 2019 - June 2020*
Science Envoy
- Selected as one of 10 graduate students from Stanford and Berkeley to communicate science to public audiences.

Stanford Diversity and First Generation Office — Stanford, CA *Oct 2018 - June 2020*
First Generation and Low Income Student Mentor
- Mentoring undergraduates from underprivileged backgrounds in all matters of student life.

Stanford Astronomical Society — Stanford, CA *Oct 2018 - Oct 2019*
Organizing Member
- Organizing and leading stargazing events for local schools and the public.

Stanford Science Pen Pals — Stanford, CA *Oct 2018 - Oct 2019*
Pen Pal
- Corresponding about science with 5th graders from a low-income school in Catalonia.

Mentoring

Jack Dinsmore (graduate) — Polarization leakage in the IXPE gas pixel detector *2022 - Mar 2023*
Josephine Wong (graduate) — Simultaneous polarization fitting of the Crab nebula and pulsar *2021 - Mar 2023*

Teaching

Stanford PHYS113 — *Computational Physics* (Lecturer and Teaching Assistant) *Winter 2021*
Stanford PHYS100 — *Introduction to Observational Astrophysics* (Teaching Assistant) *Spring 2019*

Selected Graduate Coursework

- ***APPHYS293*** (Theoretical Neuroscience)
- ***CS379C*** (Computational Models of the Neocortex)
- ***CS238*** (Decision Making Under Uncertainty)
- ***STATS207*** (Time Series)
- ***CS224N*** (Natural Language Processing)
- ***CS230*** (Deep Learning)
- ***CME212*** (Advanced Software Development)
- ***EE364a & b*** (Convex Optimization I & II)
- ***PHYS266*** (Statistical Methods in Physics)
- ***AA 214*** (Numerical Methods for Compressible Flows)
- ***CS361*** (Engineering Design Optimization)
- ***EE263*** (Linear Dynamical Systems)

Selected Publications (h-index 44)

— Machine Learning —

Benchmarking Neural Network Training Algorithms [Code]
G.E.Dahl, et al. incl. **A.L.Peirson** 2306.07179, 2023

Fishy: Layerwise Fisher Approximation for Higher-Order Neural Network Optimization
A.L.Peirson, E.Amid, Y.Chen, V.Feinberg, M.K.Warmuth, R.Anil *NeurIPS HITY workshop*, 2022

Neural Network Analysis of X-ray Polarimeter Data
A.L.Peirson, *The Handbook of X-ray and Gamma Ray Astrophysics*, *Springer Nature*, 2022

A Deep Ensemble Approach to X-ray Polarimetry
A.L.Peirson, R.W.Romani *NeurIPS ML4PS workshop*, 2021

Towards Optimal Signal Extraction for Imaging X-ray Polarimetry
A.L.Peirson, R.W.Romani. *ApJ*, 920, 40, 2021

Deep Ensemble Analysis for Imaging X-ray Polarimetry
A.L.Peirson, R.W.Romani, H.L.Marshall, J.F.Steiner, L.Baldini. *NIMA*, 986, 2021

Dank Learning: Generating Memes Using Deep Neural Networks [Code][TechCrunch][The Next Web]
A.L.Peirson & E.M.Tolunay, 1806.04510, 2018

— Astrophysics —

X-ray Polarization of BL Lacertae in Outburst
A.L.Peirson, M.Negro, I.Liodakis et al. *ApJ Letters*, 948, L25, 2023

Polarized Blazar X-rays imply Particle Acceleration in Shocks
I.Liodakis, et al. (IXPE collaboration, incl. **A.L.Peirson**) *Nature*, 611, 677, 2022

Polarized X-rays Constrain the Disk-Jet Geometry in the Black Hole X-ray Binary Cygnus X-1
H.Krawczynski, et al. (IXPE collaboration, incl. **A.L.Peirson**) *Science*, 378, 650, 2022

Testing High-Energy Emission Models for Blazars with X-ray Polarimetry
A.L.Peirson, I.Liodakis, R.W.Romani *ApJ*, 931, 59, 2022

New Tests of Millilensing in the Blazar PKS 1413+135 [Code]
A.L.Peirson, I.Liodakis, A.C.S.Readhead et al. *ApJ*, 927, 24, 2022

The Polarization Behavior of Relativistic Synchrotron Self-Compton Jets [Code]
A.L.Peirson, R.W.Romani. *ApJ*, 885, 76, 2019

Prospects for Detecting X-ray Polarization in Blazar Jets
I.Liodakis, **A.L.Peirson**, R.W.Romani. *ApJ*, 880, 29, 2019

The Polarization Behavior of Relativistic Synchrotron Jets
A.L.Peirson, R.W.Romani. *ApJ*, 864, 140, 2018